

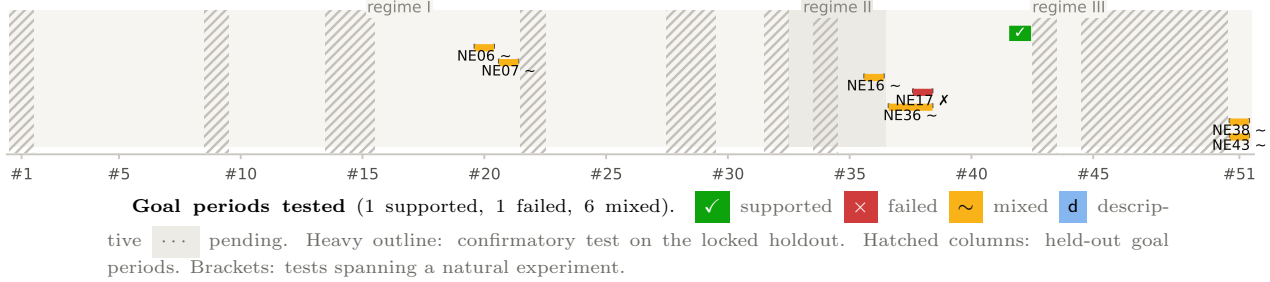
H117: J should not move when the rules do: the mid-goal reset in #32 (NE35)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1.5/5

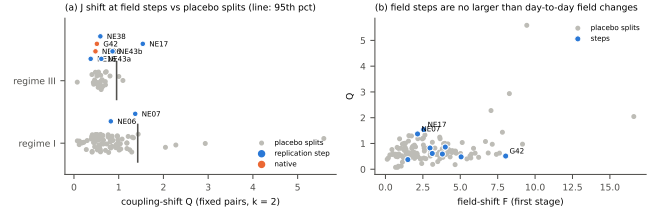
Bottom line. The fitted talk-spin coupling matrix stays at its day-to-day noise level across field steps, including the one goal switch that clearly moved the fields; most documented rule changes are not field steps in the talk data.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH355 says that a pure field change (a rule reset, a tool change, a new goal) moves each agent's field h_i but not the couplings J_{ij} . A fitted J that jumps at such a step is a field artifact. The target, NE35 in #32, is held out. H117 fits a kinetic Ising model to minute talk spins on both sides of 9 non-holdout steps (2 days each side). It compares the change in J with 109 placebo splits at ordinary day boundaries. J stays at its placebo level, but most steps are not field steps in the data. The coupling shift Q exceeds the placebo 95th percentile at 1 of 9 steps (NE17, 7 pairs, $p = 0.037$, no field shift). The field shift F beats day-to-day changes at one step only: the #41→#42 goal switch. There $Q = 0.51$ ($p = 0.74$). The holdout is not run.



Observables. (a) Coupling shift Q at the steps (blue: replication; orange: natives) against placebo splits at ordinary day boundaries (grey; line: 95th percentile), by regime. Only NE17 lies beyond it. (b) Q against the field shift F . Steps sit inside the placebo cloud on both axes; G42 (goal switch) has the largest F and a low Q .

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: a goal or rule change does not rewire who responds to whom in talk timing. At #38-type sizes, a change of ± 1.2 logistic units on half the pairs would be seen with probability 0.85, and none is seen. Talk couplings carry over a reset. Only room moves changed them, and those changes were small (switched pairs $Q = 0.93$ against fixed pairs 0.61). For an analyst: a fitted J on talk spins with block fields is not a field artifact at this resolution. Most documented rule changes, however, do not move talk fields more than an ordinary night does.

PHYSICAL MODEL

Model 02, kinetic Ising with parallel updates. A talk spin $s_i(t) = \pm 1$ marks whether agent i posted chat in minute t . Each agent's next spin is a logistic function of its own spin, the 5-min boxcar m_j of every other agent, and a per-agent field for each (day, 30-min block). The fields absorb the schedule and any field step. Under HH355, h_i moves across a step while J does not. A second estimator works on the call clock. It compares messages read just before a call (J^R) with messages posted during it and not yet read (J^U).

$$\text{logit } P(s_i^{t+1} = 1) = h_i + \delta_i(b) + K_i s_i^t + K'_i m_i^t + \sum_j J_{ij} m_j^t$$

$$Q = \langle (J_{ij}^{\text{after}} - J_{ij}^{\text{before}})^2 / (\sigma_a^2 + \sigma_b^2) \rangle_{\text{fixed pairs}}, \quad J^S = \frac{1}{2} (J_{ij} + J_{ji})$$

$$F = \langle (\bar{h}_a - \bar{h}_b)^2 / (\sigma_a^2 + \sigma_b^2) \rangle_i$$

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RESULTS VS PREDICTIONS

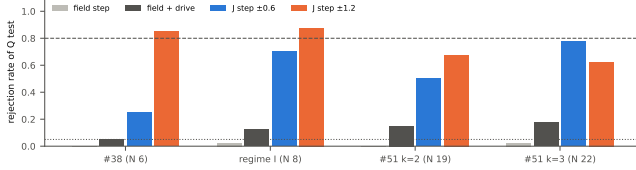
Prediction (locked)	Observed	Verdict
P0 size ≤ 0.10 ; power ≥ 0.8	size 0.00–0.03; power 0.85–0.88 at ± 1.2 , 0.25–0.78 at ± 0.6	partly
P1 Q $p > 0.05$ at $\geq 6/7$ steps	6/7 (NE17 p 0.037)	supported
P2 field step F at $\geq 4/7$	0/7 (p 0.37–0.85)	failed
P3 D_F within placebo	6/7	supported
P4 read-gated Q_R ; $J^R > J^U$	5/5; $J^R - J^U > 0$ on 9/10 sides	supported
P5 goal switches move J more	medians 0.50 vs 0.82	failed
G42 goal switch	Q 0.51 (p 0.74), F p 0.037	supported
NE38 one agent's role	its $\Delta \bar{h}$ rank 1/19, row Q rank 5/19	supported
HH kill ($\geq 2/7$ steps)	1/7	not fired

G42 ✓ supported
 NE06 ~ mixed
 NE07 ~ mixed
 NE16 ~ mixed
 NE17 × failed
 NE36 ~ mixed
 NE38 ~ mixed
 NE43 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic worlds on real step skeletons (80 null and 40 alternative worlds each). The Q test rejects a pure field step at ≤ 0.03 and a field step with a stronger common drive at 0.05–0.18. It detects rewiring of ± 1.2 on half the pairs at 0.63–0.88; at ± 0.6 only regime I reaches 0.7.

GOAL PERIODS IN DETAIL

CAVEATS

4–22 eligible agents per step, because agents with fewer than 10 talk minutes per side drop out (including both movers). The regime-III placebo pool has 26 splits, so $p \geq 0.037$. NE17's failure rests on 7 pairs. Q cannot see a global rescaling of all couplings. The naive estimator was not more fragile in the synthetic, so the field-leak rival is untested rather than rejected. The absolute J^R and J^U are confounded by call length; only their difference is read. NE35 sits on the regime I→II boundary.

ROUND 2

Run the frozen NE35/NE44 confirm after sign-off. Add log latency to the call model. Pool goal switches with fixed rooms hierarchically, using more days per side.

- **Essence.** is a fitted coupling a property of the read path or of the task? Settle it with steps that move fields far beyond nightly changes (goal switches with fixed rooms: #41 → #42, #39 → #40 within classes) and with more days per side.

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